Appendix (For Online Publication)

Social Preferences and Peer Effects in Voluntary Cooperation:
A Theoretical and Experimental Analysis

Appendix A: Experimental instructions
Appendix B: Screenshot of decision screen
Appendix C: Data overview
Appendix D: Explaining initial effort choices
Appendix E: Theoretical details

Appendix A: Instructions

[In the following we present the information subjects received during the experiment. Editorial comments like this one are added in brackets. In the majority of sessions subjects first played a three person one-shot gift-exchange game, but they did not receive any feedback about other subjects' decisions. In the sessions identified as *Experienced* subjects previously played a repeated three person gift-exchange game for eight periods in a stranger matching protocol. In this experiment they learned the prices paid by the principal, but not the choice of the co-agent. The game was presented in a buyer seller framing. Principals are buyers who offer a price. Sellers choose quality. The instructions and control questions to this first experiment can be found in the online supplement of Gächter & Thöni (2010). In the following we show the information subjects received at the beginning of the second experiment, which is the experiment we report in this paper.]

Instructions for the Second Experiment

Before we inform you about the decisions of the other two members of your group we would like to conduct a second experiment. In this second experiment you will again receive an endowment of 400 points. Your points will again be converted at the rate of:

1 Point = 3 Rappen.

The points you earn in this experiment will be paid out to you together with your earnings from the first experiment at the end of the experiment.

The Second Experiment in Detail

The second experiment is very similar to the first experiment. Again one buyer and two sellers constitute a group. The assignment of the participants to the groups is done at random. Like in the first experiment the buyer has to choose the prices for his two sellers and the sellers choose their quality. Again this experiment will be conducted only once.

However, there are three important differences relative to the first experiment:

- In this second experiment the buyers have to choose the same price for both buyers.
- The feasible prices are now 50, 100 and 200.
- The income of the buyers is calculated differently in the second experiment. Unlike in the first experiment, the sum of the qualities in the buyers' income is multiplied by 35 instead of 18. The income of the buyers is therefore calculated as:

$$\boxed{\text{Income Buyer} = 35 * (\text{Quality1} + \text{Quality2}) - \text{Price1} - \text{Price2}}$$

The calculation of the sellers' incomes remains unchanged, i.e., the calculation is the same as in the first experiment:

Income Seller = Own Price1 – 7 * (Own Quality – 1)

When deciding, you will again have access to the 'What-if-calculator' where you can check out the calculation of the incomes.

Do you have any questions?

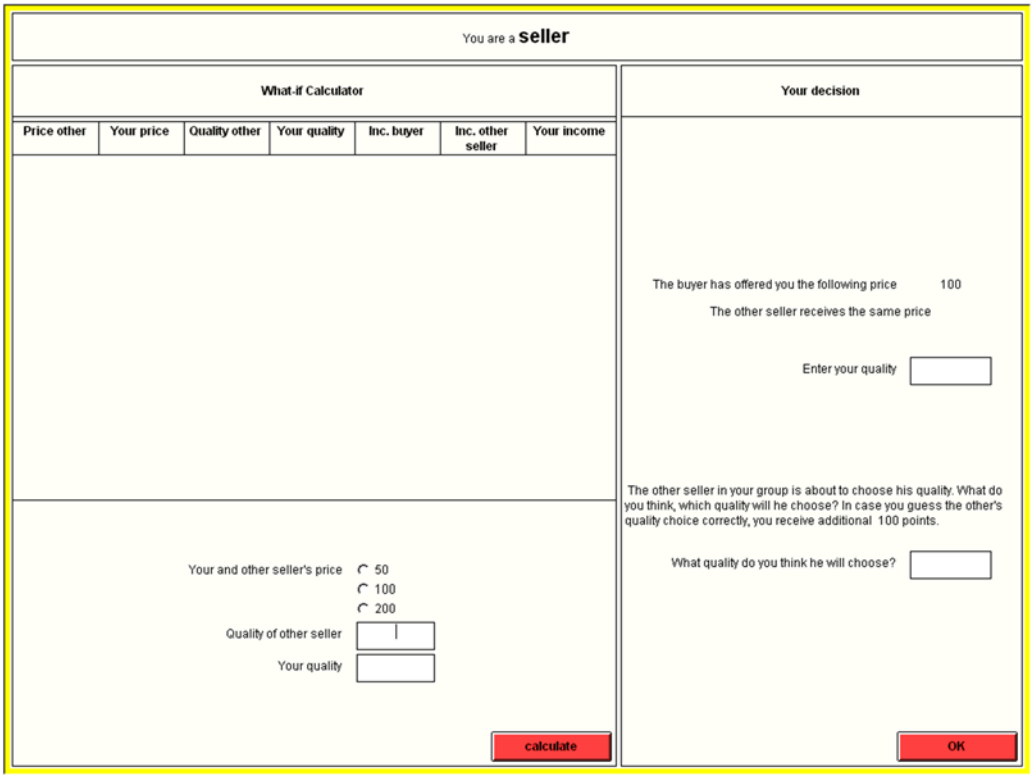
[End of the printed instructions. While the buyers choose their prices the sellers see a screen containing the following information:]

Unlike in the first experiment you will be informed about the price you receive from your buyer when deciding about your quality in this second experiment. You then can choose your quality as a number between 1 and 20. Since the buyer has to

pay the same price to both of his sellers the other seller in your group will receive the same price as you. In this moment your buyer is choosing the price. When the price is chosen you will be informed about it and you can choose your quality.

Please press “continue” when you have finished reading this information.

[After that, the sellers see the following decision screen. The lower part of the right panel was only included in the sessions where we elicited beliefs.]



[When all qualities are chosen, the sellers receive a second information screen. The shaded sentence only appears in the EIT. Everything else is equal.]

Possible revision of your quality choice

You have just chosen your quality. In the next step you can possibly revise your quality choice, if you wish to do so.

You will thereby be informed about the quality that the other seller in your group has chosen.

In the next screen you have to re-enter your quality choice. You have two possibilities: You can either leave your quality unchanged or you can change your quality. In any case you have to make an entry, even if you do not want to change your quality. In this case you can simply enter the same number as in the last screen. If you want to change your quality then enter your new quality.

Both sellers in your group will again enter a quality. However, **only one of the two sellers in your group can actually change the quality**. This means that either *your* reentered quality or the reentered quality *of the other seller* will be used for the calculation of the earnings. The computer will randomly choose one of the two sellers. For the other seller the quality choice of the last screen remains unchanged.

In case the computer chooses you, your income will be calculated with your reentered quality. On the other hand, if the computer chooses the other seller, then your quality choice from the last screen remains unchanged und will be used for the calculation of your income.

Please contact us if something is unclear. If not, please press the “continue” button to proceed to the next screen where you can choose your quality again.

[The sellers are then shown the revision screen (see Fig. 1 in the main text). In the main treatment (with wage transparency) this screen contains information about the other agent’s effort decision. Note that this is the first time the subjects learn something about the other agent’s decision. When all subjects have re-entered their quality a screen with the results of the game appears. The subjects learn the definitive qualities of the other seller and themselves and their resulting income. After that, we provide the payoff information of the first experiment.]

Appendix B: Screenshot of decision screen

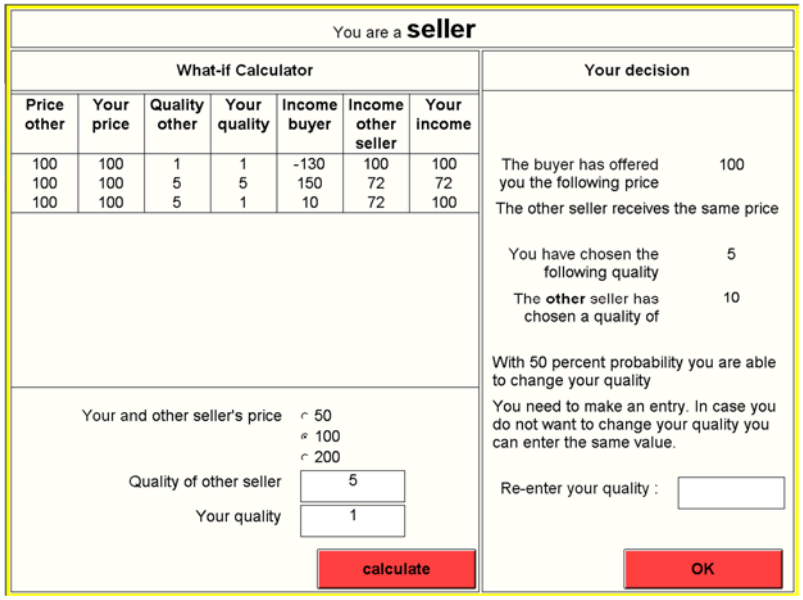


Fig. B1. Example screen shot of the decision screen at the revision stage in the Effort Information treatment.

Appendix C: Data overview

Table C1. Summary of the observations by treatment variation and contextual parameters.

	Contextual variation				Total
v	18	35	35	35	
Belief	no	no	yes	yes	
Experience	no	no	no	yes	
Main treatment: *EIT*	45	120	93	72	330
Control treatment: *NIT*	15	72	36	36	159
Total	60	192	129	108	489

Numbers indicate subjects as agents and principals. The number of agent observations is two thirds of the numbers shown.

Appendix D: Explaining initial efforts

Table D1 shows the results of a regression analysis explaining initial efforts. For this analysis we make use of Experiment 1 and 1' (without feedback, see Section 3.2). We classify our subjects into *Selfish* and *Non-selfish*. The latter are subjects who at least once chose a non-minimal effort in Experiment 1 or 1'.

We look at initial efforts in two ways: first, the likelihood of choosing a non-minimal effort and, second, initial effort levels. We apply a Probit model for the decision to choose a non-minimal initial effort and a Tobit model to investigate initial effort levels (we chose Tobit because effort is censored at 1 and 20). The independent variables are dummies for the high and low wage level; dummies that identify the contextual variations; and a dummy for experiments with belief elicitation. The baseline case is the intermediate wage and the

observations stemming from the low productivity experiments in St. Gallen. The observations from the *Experienced* sessions are clustered on matching groups.

Table D1. Probit and Tobit estimates for the initial effort decision.

	Probit: Non-minimal initial effort			Tobit: Initial effort	
	Coef	SE	ME	Coef	SE
Non-selfish (D)	1.442***	0.216	0.47	5.793***	0.947
Low wage (D)	-0.555***	0.190	-0.21	-3.002***	0.700
High wage (D)	0.379*	0.229	0.15	3.340***	0.796
Experienced (D)	-0.195	0.229	-0.08	-0.900	0.736
Belief (D)	-0.307	0.230	-0.12	-1.327	0.810
High productivity (D)	0.512*	0.310	0.19	1.429	1.300
Zurich (D)	0.065	0.278	0.03	-0.081	1.087
Constant	-1.367***	0.296		-3.565***	1.111
Σ				4.138	
N	310			310	
Log likelihood	-167.6			-479.4	
$p < \chi^2$, F	0.000			0.000	

Notes. All independent variables are dummies (D). We report coefficients and standard errors (SE) and, for the probit model, marginal effects evaluated at the mean (ME). We apply a robust estimation of the standard errors. Data from the *Experienced* treatments are clustered within matching group. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

In both models subjects classified as *Non-selfish* are significantly more likely to choose a non-minimal initial effort; they also choose higher initial effort levels than subjects classified as selfish. Effort levels increase significantly in wages; the productivity parameter v has a marginally significant effect on the probability to choose a non-minimal effort. All other variables are not significant.

Appendix E: Theoretical details

This appendix derives the reaction functions predicted by the various models of social preferences. We focus on the subgame starting after the principal has chosen the wage where both agents simultaneously choose their effort. For simplicity, we treat effort as a continuous variable in [1,20]. We use the payoff functions $x_i(w, e_i)$, and $x_P(w, e_i, e_j)$ as defined by equations (1) and (2) in the main text. By x_{ik} we denote the first derivative of $x_i()$ with respect to the kth argument. In the following we frequently need the derivatives of x_i and x_P with respect to e_i, which are $x_{i2} = -7$ and $x_{P2} = v$. Often models of social preferences predict that agent i chooses his effort such that his earnings match those of another player. We define a function $e_i = R_{i=P}(e_j)$ as the reaction function that matches agent i's and the principal's earnings whenever possible. This function is found by solving $x_i = x_P$ for e_i: [1]

[1] The notation $[x]_{[1}^{20]}$ is equivalent to $\max[\min(x, 20), 1]$.

$$e_i = R_{i=P}(e_j) = \left[\frac{3w+7-ve_j}{v+7}\right]_{[1}^{20]}. \tag{1}$$

A second important reaction function matches agent i and agent j's earnings. This is simply

$$e_i = R_{i=j}(e_j) = e_j. \tag{2}$$

Finally, in one model agent i seeks to equalize agent j's payoff and the principal's earnings. The agent i's reaction function is then

$$e_i = R_{j=P}(e_j) = \left[\frac{3w+7-(v+7)e_j}{v}\right]_{[1}^{20]}. \tag{3}$$

These three reaction functions are depicted in panel A of Fig. E1. In the intersection all three earnings are equalized. This happens, when both agents choose the effort $\bar{e} = (3w+7)/(2v+7)$.

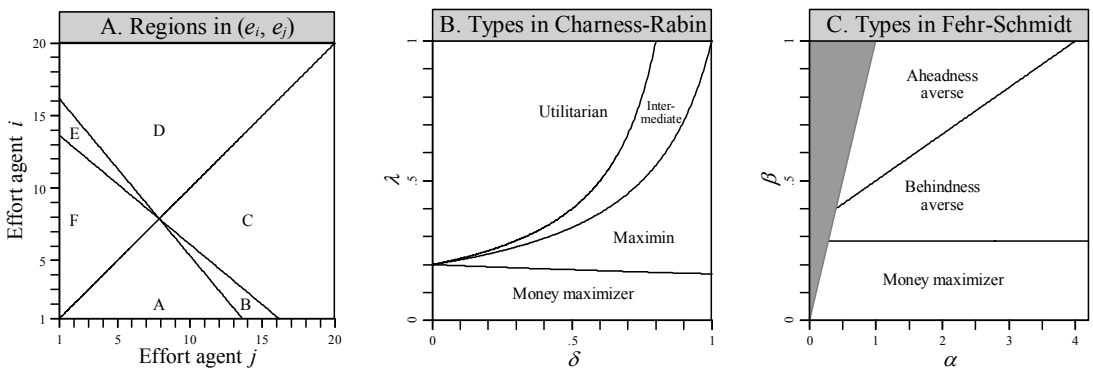


Fig. E1. Illustration of important parameter regions in the different models. Panel A: Regions in the action space of the two agents. Panel B: Parameter map and predicted types of the Charness-Rabin model. Panel C: Parameter map and predicted types for the Fehr-Schmidt model. All graphs drawn for $w = 200$ and $v = 35$.

E1. Cox et al., (2007)

Using the CES utility function from equation (3) in the main text we can derive a closed-form solution for the reaction function. The derivative of the utility function with respect to e_i gives the first order condition for interior solutions

$$\frac{\partial u_i}{\partial e_i} = \left[400 + w - 7e_i + 7\right]^{\alpha-1}(-7) + \theta_P\left[400 + v\left(e_i + e_j\right) - 2w\right]^{\alpha-1} v = 0. \tag{4}$$

Because the earnings of agent j does not depend on i's choices, θ_j drops out. Both expressions in the squared brackets are positive. Thus, in case of neutral or negative emotions towards the principal $(\theta_P \le 0)$ the derivative in (4) is negative and there is no interior solution. In this case agent i always chooses minimal effort. In case of positive emotions interior solutions exist and we can derive the reaction function R^{CFG} as

$$e_i = R^{CFG}(e_j) = \left[\frac{400 + w + 7 + \tau\left(2w - ve_j - 400\right)}{\tau v + 7} \right]_{[1}^{20]} \qquad \text{with } \tau = \left(\frac{\theta_P v}{7} \right)^{\frac{1}{\alpha - 1}}. \tag{5}$$

Since positive emotions ($\theta_P > 0$) imply $\tau > 0$, the slope of the reaction function lies in

$$-1 < \frac{\partial e_i}{\partial e_j} = -\frac{\tau v}{\tau v + 7} < 0 \tag{6}$$

for interior solutions. If $\alpha \to -\infty$ then $\tau \to 1$ and the reaction function converges to $R_{i=P}$. In Panel A of Fig. 3 in the main text we show the reaction functions for τ = .382, .442, .52, .62, .77, 1.35, 2.05, and 4.2.

E2. Charness and Rabin (*2002*)

The derivative of equation (4) in the main text with respect to e_i is

$$\frac{\partial u_i}{\partial e_i} = (1-\lambda)(-7) + \lambda\delta r + \lambda(1-\delta)(v-7) \quad \text{with } r = \begin{cases} -7 & \text{if } x_i = \min\{x_i, x_j, x_P\} \\ 0 & \text{if } x_j = \min\{x_i, x_j, x_P\} \text{ and } x_i > x_j \\ v & \text{if } x_P = \min\{x_i, x_j, x_P\} \text{ and } x_i, x_j > x_P \end{cases} \tag{7}$$

where the parameter r indicates the marginal effect of a change in e_i on the minimum income in the group. There are three cases, depending on whether agent i, j, or the principal is poorest. Agent i's effort has no marginal effect on the first-order condition but in influences r through the distribution of earnings. Whether the expression in (7) is positive or negative depends on the preference parameters λ, δ, and on r. We can calculate thresholds for the preference parameters resulting in a positive derivative:

$$\lambda > \frac{7}{\delta(7 - v + r) + v} > 0. \tag{8}$$

Note that the expression is monotonically decreasing in r. There are four different reaction functions depending on the number of inequalities that are satisfied given the three different values of r.[2]

(i) If the inequality is never satisfied, then the agent chooses minimal effort.

(ii) If only the 'weakest' inequality with $r = v$ is satisfied then agent i is ready to choose a non-minimal effort whenever the principal's income is minimal. We call this type a 'Maximin' agent (M). The reaction function for such a type is the upper boundary of the area A and F in the left panel of Fig. E1.

[2] For notational convenience we assume that the preference parameters δ and λ are drawn from a continuous density function and rule out cases where the parameters equal any of the critical values. We thereby get rid of the (not very interesting but notationally tedious) cases where the players are exactly indifferent between several effort levels within a range.

(iii) If the inequality is also satisfied for $r=0$ then the agent wants to increase his effort whenever one of the other player's income is minimal. This is the case in the areas A, B, C, and F in Fig. E1. The reaction function is the upper boundary of this area. We call this type "Intermediate" (*I*).

(iv) If the preference parameters are such that all three inequalities are satisfied then we call the agent "Utilitarian". Such an agent chooses maximum effort irrespective of what the other players do. To conclude, the reaction functions R^{CR} predicted by Charness and Rabin are either minimum or maximum effort or one of the following two:

$$e_i = R^{CR,M} = \begin{cases} R_{i=P} & \text{if } e_j \le \overline{e} \\ R_{j=P} & \text{else} \end{cases} \quad \text{and} \quad e_i = R^{CR,I} = \begin{cases} R_{i=P} & \text{if } e_j \le \overline{e} \\ R_{i=j} & \text{else} \end{cases} \tag{9}$$

Panel B of Fig. E1 shows the parameter constellations that give rise to the four types. In the Appendix of their paper Charness and Rabin enrich the model with reciprocity (discussed in the main text under 'Hybrid models'). They introduce 'demerit' parameters, in our case $d_j, d_P \in [0,1]$, indicating (inversely) how much the other player deserves to be treated nicely. The utility function is

$$\begin{aligned} u(x) = (1-\lambda)x_i + \lambda\Big[\delta \max\{x_i, x_j + bd_j, x_P + bd_P\} + \\ (1-\delta)\big(x_i + x_j[1-kd_j]_{[0} + x_P[1-kd_P]_{[0}\big) - f\left(d_j x_j + d_P x_P\right)\Big], \end{aligned} \tag{10}$$

with the nonnegative parameters b and k for the weight of the demerit parameter in the Rawlsian and the utilitarian part of the utility function. The parameter $f>0$ allows to account for destructive behavior. Note that in case there are no hard feelings with respect to the other players ($d_j, d_P = 0$) the utility function is identical to equation (4) in the main text. The derivative with respect to e_i is

$$\begin{aligned} \frac{\partial u}{\partial e_i} &= (1-\lambda)(-7) + \lambda\delta r' + \lambda(1-\delta)\left(-7 + [1-kd_P]_{[0}v\right) - fd_P v \\ \text{with } r' &= \begin{cases} v & \text{if } x_P + bd_P = \min\{x_i, x_j + bd_j, x_P + bd_P\} \\ 0 & \text{if } x_j + bd_j = \min\{x_i, x_j + bd_j, x_P + bd_P\} \\ -7 & \text{if } x_i = \min\{x_i, x_j + bd_j, x_P + bd_P\} \end{cases} \end{aligned} \tag{11}$$

The implications of the additional parameters for the reaction function are quite straightforward: For positive demerits the parameters k and f determine (in combination with λ and δ) which one of the four reaction functions is chosen. The parameter b influences the position of the reaction function. In general, a higher b shifts the reaction functions $R^{CR,M}$ and $R^{CR,I}$ downwards. If we assume $d_j=0$ (which is plausible since no information about j's actions are available) then a higher b shifts the two reaction functions downwards leaving the kink on the 45-degree line. For very high b the reaction function $R^{CR,I}$ is identical to the 45-degree line. However, none of the parameters has a marginal influence on the slope of the reaction function.

E3. Bolton & Ockenfels (2000)

Agents maximize a "motivation function" $u(x_i, \sigma_i)$, where $\sigma_i(w, e_i, e_j) = x_i(w, e_i) / X(e_i, e_j)$ is player i's share of the total earnings, with $X(e_i, e_j) = x_i + x_j + x_P$.[3] Regarding the derivatives with respect to the first and second argument the Bolton-Ockenfels model assumes $u_1 \geq 0$, $u_{11} \leq 0$, $u_2 = 0$ for $\sigma_i = 1/n$, and $u_{22} < 0$. As a result, the motivation function is strictly concave in the income share, and, for a given income x_i, it is maximized when i earns exactly the equal share, i.e., $\sigma_i = 1/n$. Agent i's first-order condition for interior solutions is:

$$\frac{\partial u}{\partial e_i} = u_1 x_{i2} + u_2 \sigma_{i2} = 0 . \tag{12}$$

To derive the slope of the reaction function we calculate the total differential of equation (12):

$$\left[u_{11} \left(x_{i2} \right)^2 + u_1 x_{i22} + u_{22} \left(\sigma_{i2} \right)^2 + u_2 \sigma_{i22} \right] de_i + \left[u_{22} \sigma_{i3} \sigma_{i2} + u_2 \sigma_{i23} \right] de_j = 0 . \tag{13}$$

From the monetary payoff functions we know that $x_{i2} = -7$, $X_1 = X_2 = (v - 7) > 0$, and all higher-order derivatives of the monetary payoff functions are zero. Furthermore, interior solutions are only possible when agent i earns at least the equal share, which implies $u_2 < 0$ (this ensures that the second-order condition is negative). The derivatives of the relative earnings function with respect to e_i (argument 2) and e_j (argument 3) are:

$$\begin{aligned}
\sigma_{i2} &= \frac{\overset{-}{x_{i2}} \overset{+}{X} - \overset{+}{x_i} \overset{+}{X_1}}{X^2} < 0 \; ; \qquad \sigma_{i3} = -\frac{\overset{+}{x_i} \overset{+}{X_2}}{X^2} < 0 \\
\sigma_{i22} &= \frac{\overset{0}{x_{i22}} \overset{+}{X^2} - 2 \overset{-}{x_{i2}} \overset{+}{X_1} \overset{+}{X} + 2 x_i \left(X_1 \right)^2 - x_i \overset{0}{X_{11}} X}{X^3} > 0 \\
\sigma_{i23} &= -\frac{\overset{-}{x_{i2}} \overset{+}{X_1} \overset{+}{X} - 2 \overset{+}{x_i} \overset{+}{X_1} \overset{+}{X_2} + x_i \overset{0}{X_{11}} X}{X^3} > 0
\end{aligned} \tag{14}$$

Hence, we can derive the slope of the reaction function:

$$\frac{de_i}{de_j} = -\frac{\overset{-}{u_{22}} \overset{-}{\sigma_{i3}} \overset{-}{\sigma_{i2}} + \overset{-}{u_2} \overset{+}{\sigma_{i23}}}{\underset{-}{u_{11}} \underset{+}{\left(x_{i2} \right)^2} + \underset{-}{u_{22}} \underset{+}{\left(\sigma_{i2} \right)^2} + \underset{-}{u_2} \underset{+}{\sigma_{i22}}} < 0 . \tag{15}$$

Furthermore, the expressions in (14) show that the effects of e_i on i's income share are stronger than the effects of e_j, i.e., $|\sigma_{i2}| > |\sigma_{i3}|$ and $\sigma_{i22} > \sigma_{i23}$. Thus, we can conclude that the slope of the reaction function must lie between 0 and –1 for interior solutions:

[3] The Bolton-Ockenfels model requires nonnegative incomes. In order to avoid negative outcomes we calculate all incomes including the endowment of 400 ECU. Overall losses are not possible in this case.

$$-1 < \frac{de_i}{de_j} < 0 \quad \text{for } 1 < e_i < 20 . \tag{16}$$

For the extreme case of a perfectly inequity-averse Bolton-Ockenfels agent we can derive a closed form of the reaction function R^{BO}. This agent would choose an effort such that, whenever possible, his earnings share equals exactly one third, i.e., $\sigma_i(w, e_i, e_j) = 1/3$. Solving for e_i gives the reaction function:

$$e_i = R^{BO}\left(e_j\right) = \left[\frac{7 + 3w - (v-7)e_j}{14 + v}\right]_{[1}^{20]} . \tag{17}$$

This is the reaction function labeled 'Exclusively equity-oriented' in Fig. 3C in the main text. For deriving the other reaction functions in this Fig. we used $u(.) = x_i - b(\sigma_i - 1/n)^2$ (as suggested by Bolton and Ockenfels (2000), p. 173) with b = 12000, 5000, 3000, 2000, 1000.

E4. Fehr & Schmidt (*1999*)

The first-order derivative of the utility function shown in equation (6) in the main text is

$$\frac{\partial u}{\partial e_i} = \begin{cases} -7 - 0.5\beta\left(-7 - \left(v+7\right)\right) & \text{if } x_i > x_j \text{ and } x_i > x_P \quad \text{(i)} \\ -7 - 0.5\alpha\left(7\right) - 0.5\beta\left(-\left(v+7\right)\right) & \text{if } x_i < x_j \text{ and } x_i > x_P \quad \text{(ii)} \\ -7 - 0.5\alpha\left(v+7\right) - 0.5\beta\left(-7\right) & \text{if } x_i > x_j \text{ and } x_i < x_P \quad \text{(iii)} \\ -7 - 0.5\alpha\left(7 + \left(v+7\right)\right) & \text{if } x_i < x_j \text{ and } x_i < x_P \quad \text{(iv)} \end{cases} \tag{18}$$

Like in Charness & Rabin (2002) e_i does not have a marginal impact on the derivatives but affects the case differentiation. Starting with case (i) where agent i earns the highest income (area A and B in the left panel of Fig. E1), the derivative is positive if $\beta > 14/(v+14)$. An agent with a lower β always chooses minimal effort. An agent with a sufficiently high β increases his effort until his income equals one of the other two players' earnings.[4]

In case (ii) agent i earns more than the principal but less than the co-agent (area F). Here we have to distinguish two cases. If the preference parameters satisfy $(v+7)\beta > 14 + 7\alpha$ then agent i is called *Aheadness averse* (*AA*). Such an agent increases her effort to adjust his payoff to the principal's earnings. In the opposite case agent i is called *Behindness averse* (*BA*), and she decreases her effort to adjust his earnings to the other agent's earnings.

In cases (iii) and (iv) (area C, D, and E) the derivatives (equation (18)) are unambiguously negative due to the parameter restriction $\alpha \geq \beta$ in the Fehr-Schmidt model.[5]

[4] Like in the case of the Charness and Rabin model we ignore the cases where parameters α and β are equal to any of the critical values.

[5] The results by Blanco, et al. (2011) suggest that this restriction might not be supported by the data. Our conclusion with regard to predicted types does, however, not crucially hinge on this assumption. As long as we assume that $\beta < 1$ no other types are predicted. In fact, we would need $\beta > 2$ to predict an additional type. This type would be identical to the intermediate type predicted by the model of Charness and Rabin (2002).

In summary, the reaction functions R^{FS} of the two Fehr-Schmidt types with interior solutions are

$$R^{FS,AA}\left(e_j\right) = R_{i=P} \qquad \text{and} \qquad R^{FS,BA}\left(e_j\right) = \begin{cases} R_{i=j} & \text{if } e_j \le \overline{e} \\ R_{i=P} & \text{else} \end{cases}. \tag{19}$$

Thus, the slope of the reaction function for interior solutions is either 1 or $-v/\left(v+7\right)$, which is in (-1,0). Panel C in Fig. E1 shows the α-β combinations that correspond to a particular type. The shaded area is excluded by the parameter restrictions.

E5. Dufwenberg & Kirchsteiger (2004)

This model measures the intentions of other players by whether their actions allow an agent to earn a high income within the possible range of earnings. The authors define an equitable income as the mean between the minimum and maximum attainable income (which might depend on beliefs about others' actions). If the actions of another player allow i to earn a payoff above (below) average, then his action is considered kind (unkind). In the three-player gift-exchange game the earnings of agent i are independent of agent j's action. Thus, there cannot be a direct effect of e_j on e_i.

However, one could suspect an indirect effect due to the fact that a reciprocal agent wants to treat the principal nicely. This is also not the case. If agent i is treated kindly by the principal, then she seeks to return the favor and allow the principal to earn a high income relative to the equitable income. Both the minimum and maximum attainable x_P depend on e_j. Thus, changes in e_j do only result in a parallel shift of the 'feasible set' of the principal's earnings, leaving the trade-off between the money maximizing and reciprocating towards the principal unchanged for agent i.

If we would incorporate the idea that agent i considers the total earnings of the principal in a way that she feels a stronger urge to reciprocate when the principal is poor then we could produce negatively sloped reaction function similar to the reaction functions derived for the models of distributional preferences.

E6. Levine (1998)

Levine's starts with distributional preferences similar to Cox et al. (2007) but with an important additional feature: the weight another player has in i's utility depends on i's estimate of j's type. Utility of an agent is written as

$$u_i = x_i + \frac{a_i + \lambda a_P}{1+\lambda} x_P + \frac{a_i + \lambda a_j}{1+\lambda} x_j\,, \tag{20}$$

with $-1 < a_i < 1$ denoting agent i's type and $0 \le \lambda \le 1$ reflecting the weight of the estimate about the other's type (a_P, a_j) in i's altruism. A player's type is drawn from a common cumulative distribution $F(a_i)$. For $\lambda = 0$ this model is equivalent to Cox et al. (2007) with

$\alpha = 1$. In case of $\lambda > 0$ players use observed behavior to update their estimate about a_P and a_j. Taking the derivative of agent i's utility gives

$$\frac{\partial u_i}{\partial e_i} = -7 + \frac{a_i + \lambda a_P}{1+n} v \,, \tag{21}$$

which is independent of e_i. Agent i provides either full or minimal effort, dependent on his own altruism parameter (a_i) and on his estimate about the principal's preferences (a_P). However, the optimization of agent i is independent of what agent j does.

E7. Falk & Fischbacher (2006)

The utility function proposed by Falk and Fischbacher has the following (simplified) form:

$$u_i = x_i + \rho_i \sum_j \vartheta_j \Delta_j \sigma_i \,, \tag{22}$$

where x_i is the monetary outcome and $\rho_i > 0$ is a preference parameter measuring the importance of reciprocal motives. For every player j the reciprocal motivation is captured by the following terms: ϑ_j is the intention factor, which is unity if player j acted intentionally and $0 \leq \varepsilon_i \leq 1$ otherwise.[6] The outcome term Δ_j measures the kindness of j towards i and the reciprocation term σ_i measures the kindness of i's reaction. In the context of the three-person gift-exchange game the utility function for agent i is:

$$\begin{aligned} u_i = x_i(w, e_i) \quad & + \rho_i \vartheta_P \left[x_i(w, e_i'') - x_P(w, e_i'', e_j'') \right] \left[x_P(w, e_i, e_j') - x_P(w, e_i'', e_j') \right] \\ & + \rho_i \vartheta_j \left[x_i(w, e_i'') - x_j(w, e_j'') \right] \left[x_j(w, e_j') - x_j(w, e_j') \right] . \end{aligned} \tag{23}$$

The first term is agent i's monetary payoff. The second term is the reciprocity term towards the principal. The outcome term (first bracket) depends on agent i's belief about the principal's belief about the two efforts. The first expression in brackets shows the agent's belief about the principal's kindness. This expression depends on the agent's beliefs about the principal's beliefs about the two efforts. These second-order beliefs are denoted as e_i'' and e_j''. If this term is positive, agent i perceives the principal's actions as kind. The second expression in brackets shows the reciprocal reaction, that is, the influence of agent i's effort on the principal's payoff (given agent i's belief about the other agent's effort e_j'). The third term shows the reciprocity towards the other agent. Since the agents cannot influence each others' payoffs the third term is independent of e_i and therefore irrelevant for agent i's optimization. If we differentiate the utility function (23) with respect to e_i we can write the first order condition as:

$$\frac{\partial u_i}{\partial e_i} = -7 + \rho_i \vartheta_P \left[w - 7(e_i'' - 1) - \left\{ v(e_i'' + e_j'') - 2w \right\} \right] v = 0 \,. \tag{24}$$

[6] The parameter ε_i is a second preference parameter that measures a player's pure concern for an equitable outcome. Thereby, $\varepsilon_i = 1$ means that intentions do not matter while $\varepsilon_i = 0$ describes the case where the other agent's income is only taken into account if his actions are intentional.

In equilibrium, beliefs must be consistent, that is, $e_i = e_i' = e_i''$ and $e_j = e_j' = e_j''$. From the first-order condition we can therefore derive agent *i*'s reaction function R^{FF} as

$$e_i = R^{FF}(e_j) = \left[\frac{1}{v+7} \left(3w + 7 - \frac{7}{v\rho_i \vartheta_P} - ve_j \right) \right]_{[1}^{20]} . \tag{25}$$

Thus, the Falk-Fischbacher model predicts a reaction function that is linear in the other agent's effort with a slope of $-1 < \partial e_i / \partial e_j = -v/(v+7) < 0$. For interior solutions the slope is independent of the preference parameter ρ_i and the intention factor ϑ_F. The intention factor is equal to one if the principal pays a wage of 100 or 200. This is an intentionally kind act, since it could have made the agent worse off by paying a wage of 50. Being paid a wage of 50, on the other hand, is perceived as non-intentional and therefore we set $\vartheta_P = \varepsilon_i$. The preference parameter ρ_i has a very straightforward influence on the reaction function: no concern for reciprocity $(\rho_i \to 0)$ leads to minimal effort, a higher concern for reciprocity shifts the reaction function upwards. The limit case $(\rho_i \to \infty)$ is identical to $R_{i=P}(e_j)$.

E8. López-Pérez (2008)

This model formalizes norm abiding preferences. The underlying idea is simple: there is a behavioral norm which norm abiding people like to follow, called the E-norm (E stands for both efficiency and equity). This norm is determined by the so-called *fairmax* distribution of the monetary payoffs, which is the result of the maximization of the total earnings minus the difference between the most and the least well-off player:

$$\max_{e_i, e_j, w} \quad F = x_i + x_j + x_P - \delta(\max[x_i, x_j, x_P] - \min[x_i, x_j, x_P]), \tag{26}$$

where $1 > \delta > 0$ measures the importance of the concern for inequality relative to efficiency. If δ is small then efficiency dominates and both efforts need to be maximal to maximize the sum of the earnings. In addition, wages must be maximal to minimize the income differences. If δ is close to one then the inequality part dominates and the unique solution of the maximization problem is that the principal pays the highest wage and the agents choose their efforts to equalize all three earnings, $\bar{e}\big|_{w=200} = 607/(2v+7)$. Starting from a situation with maximal efforts the inequality is most efficiently reduced when both efforts are lowered by the same amount. The critical δ is where the marginal benefit (in terms of F) equals the marginal cost. The marginal benefit of increasing both efforts is $2(v-7)$, the marginal cost, if efforts go beyond the point where earnings are equalized, is $\delta(2v-(-7))$, which leads to $\delta = 8/11$. Ignoring the case where δ equals the critical values we have a unique E-norm in this game, characterized by the following actions:

$$\{\tilde{e}, \tilde{w}\} = \begin{cases} \{20, 200\} & \text{if } \delta < 8/11 \\ \{\bar{e}\big|_{w=200}, 200\} & \text{if } \delta > 8/11 \end{cases} \tag{27}$$

Whether a player sticks to the norm or not depends on a preference parameter $\gamma > 0$ and the observed behavior of other players. For simplicity we assume homogeneous players. Denote by r the number of players who follow the norm or have made no choice so far. An agent's utility is given by

$$u_i = \begin{cases} x_i & \text{if } e_i = \tilde{e} \\ x_i - \gamma r & \text{if } e_i \neq \tilde{e} \end{cases} \tag{28}$$

The number r is updated during the game and γr measures the non-pecuniary cost of norm deviation, which is increasing in the number of norm followers in the group. From the utility function it is clear that an agent does only have to consider two actions: if the cost of deviating from the norm are sufficiently high the agent chooses $e_i = \tilde{e}$, else she deviates from the norm and chooses $e_i = 1$ to maximize his earnings. Apart from the preference parameter γ two factors influence the decision. When the game starts we have $r = 3$, because no player has taken an action so far. If an agent observes that the principal does not pay the highest wage and/or the other agent does not follow the norm then r is reduced by one or two units.

Let k be the monetary cost of following the norm, which is the cost of the effort provided. Depending on δ, we have $k = c(\bar{e}\,|_{w=200})$ or $k = c(20)$. An agent's reaction function is

$$R_i(e_j) = \begin{cases} 1 & \text{if } k > \gamma r \\ \tilde{e}_i & \text{if } k \leq \gamma r \end{cases} \tag{29}$$

An agent with $3\gamma > k$ initially chooses $e_i = \tilde{e}_i$ if the principal pays the high wage (otherwise the condition is $2\gamma > k$). If, in the revision stage, the agent learns that the co-agent did not follow the norm ($e_j = 1$), then the agent will revise the own effort to $e_i = 1$ if γ satisfies $3\gamma > k > 2\gamma$ (or $2\gamma > k > \gamma$ in case of $w < 200$). Otherwise she will stick to the norm.

E9. Ellingsen & Johannesson (2008)

Ellingson and Johannesson (2008, p. 1003) describe a separating equilibrium for the two-person gift-exchange game. We adapt their model to our case and focus only on the agent's problem. There are two types of players, altruistic (θ_H) and selfish (θ_L) players, with $0 \leq \theta_L < \theta_H < 7/v$. The upper bound on altruism ensures that no agent wants to choose non-minimal effort in the absence of pride concerns. Dependent on the type of the other player there are two levels of salience of the other player's esteem, σ_H und σ_L, with $\sigma_H > \sigma_L$. We only look at cases where the principal paid the highest wage and has a high salience for the two agents. There is a common prior $0 < p < 1$ that denotes the probability of an agent being altruistic. We define $v_i(e_i, e_j) \equiv x_i + \theta_i(x_j + x_P)$ as the material part of agent i's utility. The incentive compatibility constraint for a selfish agent is then

$$v_i(1, e_j) + \left(\sigma_H + p\sigma_H + (1-p)\sigma_L\right)\theta_L = v_i(\tilde{e}_i, e_j) + \left(\sigma_H + p\sigma_H + (1-p)\sigma_L\right)\theta_H . \tag{30}$$

The left-hand expression is i's utility if she chooses minimal effort and gets low esteem from both other players. Esteem from the principal is weighed by σ_H; for the other agent's type is not known it is the expected weight. The right-hand side is i's utility of choosing a non-minimal effort $\tilde{e}_i$, assuming this leads to high esteem. This effort is:

$$\tilde{e}_i(p) = \frac{\left(\sigma_H + p\sigma_H + (1-p)\sigma_L\right)\left(\theta_H - \theta_L\right)}{7 - \theta_L v} + 1 \tag{31}$$

In a separating equilibrium the altruistic agent chooses a non-minimal effort of $e_i = \tilde{e}_i$ to credibly signal his altruism to the other two players; the selfish agent chooses $e_i = 1$. What changes in the revision stage when the agents choose the revised effort $\hat{e}_i$? When the co-agent's effort is disclosed the agent knows whether the co-agent is altruistic or selfish. This changes the effort which credibly signals altruism. We can calculate the effort by setting either $p = 0$ or $p = 1$. Because the expression in (31) is increasing in p we have $\tilde{e}_i(1) > \tilde{e}_i(p) > \tilde{e}_i(0)$. Thus, a selfish agent always chooses minimal effort. An altruistic agent initially chooses $e_i = \tilde{e}_i(p)$ and revises upwards to $\hat{e}_i = \tilde{e}_i(1)$ if the co-agent chose $e_j = \tilde{e}_i(p)$ and downwards to $\hat{e}_i = \tilde{e}_i(0)$ if the co-agent chose a minimal effort.

References

Blanco, M., Engelmann, D., & Normann, H. T. (2011). A within-subject analysis of other-regarding preferences. *Games and Economic Behavior, 72*, 321-338.

Bolton, G. E., & Ockenfels, A. (2000). Erc: A theory of equity, reciprocity, and competition. *American Economic Review, 90*, 166-193.

Charness, G., & Rabin, M. (2002). Understanding social preferences with simple tests. *Quarterly Journal of Economics, 117*, 817-869.

Cox, J. C., Friedman, D., & Gjerstad, S. (2007). A tractable model of reciprocity and fairness. *Games and Economic Behavior, 59*, 17-45.

Dufwenberg, M., & Kirchsteiger, G. (2004). A theory of sequential reciprocity. *Games and Economic Behavior, 47*, 268-298.

Ellingsen, T., & Johannesson, M. (2008). Pride and prejudice: The human side of incentive theory. *American Economic Review, 98*, 990-1008.

Falk, A., & Fischbacher, U. (2006). A theory of reciprocity. *Games and Economic Behavior, 54*, 293-315.

Fehr, E., & Schmidt, K. M. (1999). A theory of fairness, competition, and cooperation. *Quarterly Journal of Economics, 114*, 817-868.

Gächter, S., & Thöni, C. (2010). Social comparison and performance: Experimental evidence on the fair wage-effort hypothesis. *Journal of Economic Behavior & Organization, 76*, 531-543.

Levine, D. K. (1998). Modeling altruism and spitefulness in experiments. *Review of Economic Dynamics, 1*, 593-622.

López-Pérez, R. (2008). Aversion to norm-breaking: A model. *Games and Economic Behavior, 64*, 237-267.